# MATH I - Week 5

Fall 2026

In the following, you will see solutions to the exercises worked out in class. They are from textbook's Section 2.4 (on Continuity).

### NOTE: Please, pay attention to the exercises with "removable disconinuities" to better understand this type of discontinuities.

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## SECTION 2.4 EXERCISES

*For the following exercises, determine the point(s), if any, at which each function is discontinuous. Classify any discontinuity as jump, removable, infinite, or other.*

**131.** $f(x) = \frac{1}{\sqrt{x}}$

**132.** $f(x) = \frac{2}{x^2+1}$

**133.** $f(x) = \frac{x}{x^2-x}$

**134.** $g(t) = t^{-1} + 1$

**135.** $f(x) = \frac{5}{e^x-2}$

**136.** $f(x) = \frac{|x-2|}{x-2}$

**137.** $H(x) = \tan 2x$

**138.** $f(t) = \frac{t+3}{t^2+5t+6}$

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# SOLUTIONS

We determine discontinuities by checking where the functions are undefined and evaluating the corresponding limits.

**131.** $f(x) = \dfrac{1}{\sqrt{x}}$

For a real-valued function, the square root requires $x \geq 0$, but the denominator cannot equal zero. Hence,

$$\text{Dom}(f) = (0, \infty).$$

**Note:** $\text{Dom}(f) = (0, \infty)$ means that the domain $(\text{Dom})$ of the function $f(x)$ is $(0, \infty)$.

As $x \to 0^+$,

$$\frac{1}{\sqrt{x}} \to +\infty.$$

Therefore, $x = 0$ is a vertical asymptote.

$$\boxed{x = 0 : \text{ infinite discontinuity}}$$

The function is continuous everywhere on its domain $(0, \infty)$.

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**132.** $f(x) = \dfrac{2}{x^2+1}$

Since

$$x^2 + 1 \geq 1 > 0$$

for every real $x$, the denominator never equals zero. A rational function is continuous wherever its denominator is nonzero.

Therefore,

$$\boxed{f \text{ is continuous for every } x \in \mathbb{R}.}$$

There are no discontinuities.

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**133.** $f(x) = \dfrac{x}{x^2-x}$

Factor the denominator:

$$x^2 - x = x(x-1).$$

Therefore,

$$f(x) = \tfrac{x}{x(x-1)} = \tfrac{1}{x-1}, \qquad x \neq 0, 1.$$

The original function is undefined at $x = 0$ and $x = 1$.

**At** $x = 0$

The factor $x$ cancels, and

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \tfrac{1}{x-1} = -1.$$

The limit exists and is finite, but $f(0)$ is undefined. Thus, $x = 0$ is a removable discontinuity.

**At** $x = 1$

Using the simplified expression,

$$f(x) = \tfrac{1}{x-1}.$$

Then

$$\lim_{x\to 1^-} f(x) = -\infty, \qquad \lim_{x\to 1^+} f(x) = +\infty.$$

Thus, $x = 1$ is an infinite discontinuity.

$$\boxed{x = 0 : \text{ removable discontinuity}}$$

$$\boxed{x = 1 : \text{ infinite discontinuity}}$$

The removable discontinuity could be repaired by defining

$$f(0) = -1.$$

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**134.** $g(t) = t^{-1} + 1$

Rewrite the function as

$$g(t) = \frac{1}{t} + 1.$$

It is undefined at $t = 0$. Moreover,

$$\lim_{t \to 0^-} \left(\frac{1}{t} + 1\right) = -\infty,$$

while

$$\lim_{t \to 0^+} \left(\frac{1}{t} + 1\right) = +\infty.$$

Therefore, $t = 0$ is a vertical asymptote.

$$\boxed{t = 0 : \text{ infinite discontinuity}}$$

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**135.** $f(x) = \dfrac{5}{e^x - 2}$

The denominator is zero when

$$e^x - 2 = 0.$$

Thus,

$$e^x = 2$$

and

$$x = \ln 2.$$

As $x \to (\ln 2)^-$, we have $e^x - 2 \to 0^-$, so

$$\lim_{x \to (\ln 2)^-} \frac{5}{e^x - 2} = -\infty.$$

As $x \to (\ln 2)^+$, we have $e^x - 2 \to 0^+$, so

$$\lim_{x \to (\ln 2)^+} \frac{5}{e^x - 2} = +\infty.$$

Therefore,

$$\boxed{x = \ln 2 : \text{ infinite discontinuity}}$$

and $x = \ln 2$ is a vertical asymptote.

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**136.** $f(x) = \dfrac{|x - 2|}{x - 2}$

The function is undefined at $x = 2$.

Recall that

$$|x - 2| = \begin{cases} -(x - 2), & x < 2, \\ x - 2, & x > 2. \end{cases}$$

Therefore,

$$f(x) = \begin{cases} -1, & x < 2, \\ 1, & x > 2. \end{cases}$$

The one-sided limits are

$$\lim_{x \to 2^-} f(x) = -1$$

and

$$\lim_{x \to 2^+} f(x) = 1.$$

Because these limits are different, the two-sided limit does not exist.

$$\boxed{x = 2 : \text{ jump discontinuity}}$$

Defining a value at $x = 2$ cannot remove the discontinuity because the one-sided limits are unequal.

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**137.** $H(x) = \tan(2x)$

Recall that

$$\tan(2x) = \frac{\sin(2x)}{\cos(2x)}.$$

The function is undefined wherever

$$\cos(2x) = 0.$$

This occurs when

$$2x = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

Dividing by $2$,

$$x = \frac{\pi}{4} + \frac{k\pi}{2}.$$

At each of these points, the tangent function becomes unbounded and has a vertical asymptote.

Therefore,

$$\boxed{x = \frac{\pi}{4} + \frac{k\pi}{2}, \quad k \in \mathbb{Z} : \text{ infinite discontinuities.}}$$

Some of these points are

$$\dots, -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}, \dots$$

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**138.** $f(t) = \dfrac{t+3}{t^2+5t+6}$

Factor the denominator:

$$t^2 + 5t + 6 = (t+2)(t+3).$$

Thus,

$$f(t) = \frac{t+3}{(t+2)(t+3)} = \frac{1}{t+2}, \qquad t \neq -3, -2.$$

The original function is undefined at $t = -3$ and $t = -2$.

**At $t = -3$**

The factor $t + 3$ cancels, so

$$\lim_{t\to -3} f(t) = \lim_{t\to -3} \frac{1}{t+2} = -1.$$

The limit exists, but the original function is undefined. Therefore,

$$\boxed{t = -3 : \text{ removable discontinuity}}.$$

The hole could be repaired by defining

$$f(-3) = -1.$$

**At $t = -2$**

The remaining denominator becomes zero:

$$\frac{1}{t+2}.$$

Furthermore,

$$\lim_{t\to -2^-} f(t) = -\infty, \qquad \lim_{t\to -2^+} f(t) = +\infty.$$

Therefore,

$$\boxed{t = -2 : \text{ infinite discontinuity}}.$$

**Summary**

| Exercise | Discontinuity point(s) | Classification |
|---|---|---|
| 131 | $x = 0$ | Infinite |
| 132 | None | Continuous on $\mathbb{R}$ |
| 133 | $x = 0$ | Removable |
| | $x = 1$ | Infinite |
| 134 | $t = 0$ | Infinite |
| 135 | $x = \ln 2$ | Infinite |
| 136 | $x = 2$ | Jump |
| 137 | $x = \frac{\pi}{4} + \frac{k\pi}{2}$ | Infinite |
| 138 | $t = -3$ | Removable |
| | $t = -2$ | Infinite |

# Another set of problems (you may skip exercise 146)

*In the following exercises, find the value(s) of k that makes each function continuous over the given interval.*

**145.** $f(x) = \begin{cases} 3x+2, & x < k \\ 2x-3, & k \le x \le 8 \end{cases}$

**146.** $f(\theta) = \begin{cases} \sin\theta, & 0 \le \theta < \frac{\pi}{2} \\ \cos(\theta + k), & \frac{\pi}{2} \le \theta \le \pi \end{cases}$

**147.** $f(x) = \begin{cases} \frac{x^2+3x+2}{x+2}, & x \neq -2 \\ k, & x = -2 \end{cases}$

**148.** $f(x) = \begin{cases} e^{kx}, & 0 \le x < 4 \\ x+3, & 4 \le x \le 8 \end{cases}$

**149.** $f(x) = \begin{cases} \sqrt{kx}, & 0 \le x \le 3 \\ x+1, & 3 < x \le 10 \end{cases}$

# SOLUTIONS:

For each piecewise function, the individual formulas are continuous on their respective intervals. Therefore, we only need to check the point where the formula changes.

**145.**

$$f(x) = \begin{cases} 3x+2, & x < k, \\ 2x-3, & k \le x \le 8. \end{cases}$$

The possible discontinuity occurs at (x=k).

The left-hand limit is

$$\lim_{x \to k^-} f(x) = 3k + 2.$$

From the second branch,

$$f(k) = 2k - 3,$$

and

$$\lim_{x \to k^+} f(x) = 2k - 3.$$

For continuity,

$$3k + 2 = 2k - 3.$$

Solving,

$$k = -5.$$

Therefore,

$$\boxed{k = -5}.$$

Checking:

$$3(-5) + 2 = -13$$

and

$$2(-5) - 3 = -13.$$

Thus, both pieces meet at $(-5, -13)$.

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**146.**

$$f(\theta) = \begin{cases} \sin\theta, & 0 \le \theta < \dfrac{\pi}{2}, \\ \cos(\theta + k), & \dfrac{\pi}{2} \le \theta \le \pi. \end{cases}$$

The only possible discontinuity is at

$\theta = \frac{\pi}{2}$.

The left-hand limit is

$\lim_{\theta \to (\pi/2)^-} \sin\theta = \sin\frac{\pi}{2} = 1.$

The function value from the second branch is

$f\left(\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2} + k\right).$

Continuity requires

$\cos\left(\frac{\pi}{2} + k\right) = 1.$

The cosine equals (1) when its argument is an integer multiple of $2\pi$:

$\frac{\pi}{2} + k = 2\pi n, \qquad n \in \mathbb{Z}.$

Therefore,

$$\boxed{k = 2\pi n - \frac{\pi}{2}, \qquad n \in \mathbb{Z}.}$$

Equivalently,

$$\boxed{k = -\frac{\pi}{2} + 2\pi n, \qquad n \in \mathbb{Z}.}$$

Examples include

$k = -\frac{\pi}{2}, \quad \frac{3\pi}{2}, \quad -\frac{5\pi}{2}, \ldots$

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**147.**

$$f(x) = \begin{cases} \dfrac{x^2 + 3x + 2}{x + 2}, & x \neq -2, \\ k, & x = -2. \end{cases}$$

Factor the numerator:

$x^2 + 3x + 2 = (x + 1)(x + 2).$

For $x \neq -2$,

$f(x) = \frac{(x+1)(x+2)}{x+2} = x + 1.$

Therefore,

$\lim_{x \to -2} f(x) = \lim_{x \to -2}(x + 1) = -1.$

For continuity, the function value at $x = -2$ must equal this limit:

$k = -1.$

Hence,

$\boxed{k = -1}.$

This fills the removable hole in the original rational expression.

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**148.**

$$f(x) = \begin{cases} e^{kx}, & 0 \le x \le 4, \\ x + 3, & 4 \le x \le 8. \end{cases}$$

The two branches meet at $x = 4.$

From the first branch,

$f(4) = e^{4k}.$

From the second branch,

$f(4) = 4 + 3 = 7.$

For the function to be well-defined and continuous at $x = 4$, the two values must be equal:

$e^{4k} = 7.$

Take natural logarithms:

$4k = \ln 7.$

Therefore,

$\boxed{k = \dfrac{\ln 7}{4}}.$

Numerically,

$k \approx \frac{1.94591}{4} \approx 0.4865.$

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**149.**

$$f(x) = \begin{cases} \sqrt{kx}, & 0 \le x \le 3, \\ x + 1, & 3 < x \le 10. \end{cases}$$

The possible discontinuity occurs at $x = 3.$

From the first branch,

$f(3) = \sqrt{3k}.$

The right-hand limit from the second branch is

$\lim_{x \to 3^+} (x + 1) = 4.$

Continuity requires

$\sqrt{3k} = 4.$

Squaring both sides gives

$$3k = 16,$$

so

$$\boxed{k = \frac{16}{3}}.$$

This value is positive, so $\sqrt{kx}$ is real-valued for every $x \in [0, 3]$.

Checking:

$$f(3) = \sqrt{3 \cdot \tfrac{16}{3}} = \sqrt{16} = 4,$$

which agrees with the right-hand limit.

### Summary

| Exercise | Continuity condition | Value(s) of $k$ |
|---|---|---|
| 145 | $3k + 2 = 2k - 3$ | $\boxed{-5}$ |
| 146 | $\cos(\pi/2 + k) = 1$ | $\boxed{2\pi n - \pi/2,\ n \in \mathbb{Z}}$ |
| 147 | $k = \lim_{x \to -2} \frac{x^2 + 3x + 2}{x + 2}$ | $\boxed{-1}$ |
| 148 | $e^{4k} = 7$ | $\boxed{\ln(7)/4}$ |
| 149 | $\sqrt{3k} = 4$ | $\boxed{16/3}$ |

In [ ]:

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